

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination May 2017

PH121Pharmaceutical Biology-II

Date: 10.05.2017, Wednesday Time:10:00a.m. to1:00p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. **SECTION – I** comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to **SECTION – I** are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for **SECTION – I** is 30 minutes. Answers to **SECTION – I** must be written during the first 30 minutes of the examination.
3. Answers to **SECTION – II** and **SECTION – III** are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

- | | | |
|------------|---|-----------|
| Q 1 | Attempt all questions. Each question is of one mark. | 20 |
| 1. | One of the following vitamin is synthesized by microbes
[A]Vitamin K
[B]Vitamin D
[C]Vitamin A
[D]Vitamin C | |
| 2. | Exogenous antigens process. antigen binds with
[A]MHC-II molecule
[B]MHC-I molecule
[C]Both
[D]None | |
| 3. | Which ion plays a key role in establishing the resting membrane potential and in the repolarization phase of action potentials in neurons and muscle fibers?
[A] Calcium
[B] Magnesium
[C] Chloride
[D] Potassium | |
| 4. | Chyle is
[A]Lymph with protein
[B] Lymph with lipids
[C] Lymph with carbohydrate
[D] Lymph with blood | |
| 5. | Vasopressin hormone is also known as
[A]Aldosterone
[B]Antidiuretic hormone
[C]Angiotensin-II
[D]Atrial Natriuretic Peptide | |
| 6. | Primary lymphatic organ is,
[A]Lymph nodes
[B]Spleen
[C]Red bone marrow
[D]Lymphatic nodules | |
| 7. | Deficiency of Vitamin B causes | |

- [A] Osteomalacia
 - [B] Atherosclerosis
 - [C] Beriberi
 - [D] Scurvy
8. Genetic engineering primarily involves the manipulation of genetic material
- [A] Protein
 - [B] Chromosome
 - [C] DNA
 - [D] RNA
9. Which one is fat soluble vitamin?
- [A] Vitamin H
 - [B] Vitamin A
 - [C] Vitamin B12
 - [D] Vitamin C
10. When injury occurs, mast cell release _____
- [A] Prostaglandin
 - [B] Histamine
 - [C] Leukotrine
 - [D] Kinin
11. In antibody-mediated immunity, B cells transform into
- [A] Neutrophil
 - [B] Mast cell
 - [C] Plasma cells
 - [D] Cytotoxic T cell
12. Most abundant cation in extracellular fluid is
- [A] Na^+
 - [B] K^+
 - [C] Ca^+
 - [D] Mg^+
13. Following is not a stop codon in translation
- [A] UAA
 - [B] UAG
 - [C] UGA
 - [D] UGG
14. Cytotoxic T cells kill the antigen by granzyme which causes
- [A] Apoptosis
 - [B] Cytolysis
 - [C] Chemotaxis
 - [D] All of above
15. When water loss is greater than water gain, it causes
- [A] Sweating
 - [B] Dehydration
 - [C] Diarrhea
 - [D] All of above
16. Source of vitamin D is
- [A] Green vegetables
 - [B] Fish oil
 - [C] Microbes
 - [D] UV rays
17. Hormones that regulate renal Na and Cl reabsorption are
- [A] Angiotensin-II
 - [B] Follicle stimulating hormone
 - [C] Leutilizing hormone
 - [D] Progesterone

18. Rickets occurs because of deficiency of
 [A]Retinol
 [B]Biotin
 [C]Calciferol
 [D]Ascorbic acid
19. Most abundant anion in intracellular fluid is
 [A] SO_4^-
 [B] Cl^-
 [C] HPO_4^{2-}
 [D] HCO_3^-
20. Deficiency of Vitamin A causes
 [A]Osteoporosis
 [B]Xerophthalmia
 [C]Scurvy
 [D]Rickets

SECTION – II

- Q 2** Attempt any TWO of the following
- A** Define and classify vitamins. Prepare a note on vitamin C 05
- B** What is inflammation? Prepare a note on different stages of inflammation. 05
- C** Define and explain Innate Immunity. 05
- Q 3** Attempt any TWO of the following 05
- A** Apply a pathway for antigen processing which is in inside cell 05
- B** Give an account of the water distribution and its balance in the body. 05
- C** Describe the basic principles underlying the recombinant DNA technology. 05

SECTION – III

- Q.4** Attempt any FOUR of the following
- A** Write a note on DNA replication. 05
- B** Prepare a note on Nucleotide. 05
- C** Define: i) Gene, ii) Genetic, iii) Genome, iv) Biotechnology, v) Chromosomes. 05
- D** Write a note on Recombinant vaccines. 05
- E** Write a short note on Applications of Biotechnology Molecular diagnosis of genetic diseases. 05
- F** Define Lymphadenopathy. Describe composition and function of Lymph. 05
- Q 5** Attempt any FOUR of the following.
- A** Describe the functions, daily requirements, sources and deficiency manifestations of vitamin A. 05
- B** Write a note on Translation. 05
- C** Discuss anatomy and function of Thymus. 05
- D** Compare the composition of electrolytes in the extracellular and intercellular fluids regulation of electrolyte balance. 05
- E** Differentiate between Cell mediated and Humoral Immunity. 05
- F** Discuss about mode of transmission and preventive measures of Tuberculosis. 05